

Elementary Matrices

* Lemma 1.1 Matrix multiplication is associative i.e. $A.(B.C) = (A.B).C$

* Lemma 1.2 If $P = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $Q = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}$ are matrices s.t. $\exists R = P.Q$ then $R = \begin{bmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{bmatrix}$
if # columns of A and C = # rows of A' and B'

* Defn 1.3 A matrix A is called invertible if $\exists B$ s.t. $A.B = B.A = I$

* Lemma 1.4 A is invertible $\iff A^T$ is invertible

* Lemma 1.5 If A, B and C are matrices s.t. A and B are invertible and $C = A.B \implies C$ is invertible

* Lemma 1.6 Elementary Matrices $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$
 $\begin{cases} \longrightarrow \text{Effect of left and right multiplication} \\ \longrightarrow \text{They are invertible} \end{cases}$

* Defn 1.7 Reduced row echelon form (rref):

- ① The first non zero element in a row, i.e. pivot, is 1. All the elements above and below the pivot are zero
- ② If row i and row j are two rows of rref s.t. $j > i$, then if both rows have pivot then pivot of row j is to the left of row i
- ③ If row i is all zero then all the rows below it are all zero.

* Lemma 1.8

A (square matrix)
 $\swarrow$ invertible $\searrow$ non invertible
 $\swarrow$ $\text{rref}(A) = I$ $\searrow$ $\text{rref}(A) = \begin{bmatrix} P & Q \\ 0 & 0 \end{bmatrix}$ and also $\exists$ invertible matrix M s.t. $\text{rref}(A) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} M$

* Theorem 1.9

If A, B, C are matrices s.t. $C = A.B$
 C is invertible $\iff$ Both A and B are invertible

Hint Use 1.8 and 1.5

* Lemma 1.10

A is an invertible matrix $\iff A$ is product of elementary matrices

* Defn 1.11

Determinant : $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ s.t

① $\det(I) = 1$

② $\det \begin{pmatrix} \vdots & \vdots & \vdots \\ a \vdots + b \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix} = a \det \begin{pmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix} + b \det \begin{pmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix}$, i.e. det is linear in rows

③ If two rows of A are same then $\det(A) = 0$

* Lemma 1.12

If E is an elementary matrix and A are square matrices then

$$\det(EA) = \det(E) \det(A)$$

* Theorem 1.13

If A and B are square matrices s.t $\exists A, B$ then

$$\det(AB) = \det(A) \det(B)$$

Hint 1.12, 1.10

* Lemma 1.14

If A is a square matrix then $\det(A) = \det(A^T)$

* Theorem 1.15

If A, P, Q are square matrices s.t $A = \begin{bmatrix} P & 0 \\ 0 & Q \end{bmatrix}$ then $\det(A) = \det(P) \det(Q)$

Hint (i) P or Q is non invertible

(ii) P and Q are invertible

$$f(B) = \frac{\det \begin{bmatrix} B & 0 \\ 0 & Q \end{bmatrix}}{\det(Q)}$$

$f(B)$ satisfies 1.11 and $\exists$ a det map.

* Note 1.16

Discussions upto here can be extended to complex matrices

* Defn 1.17

S_n : Symmetric group: Group of permutations of n elements

Cyclic notation for elements of S_n

* Defn 1.18

GL_n : General linear group

$$GL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } \exists B \text{ s.t. } AB = BA = I \}$$

* Defn 1.19

SL_n : Special linear group

$$SL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } AB = BA = I, \det(A) = 1 \}$$

* Defn 1.20

O_n : Orthogonal group

$$O_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } A \cdot A^T = A^T \cdot A = I \}$$

* Defn 1.21

SO_n : Special Orthogonal group

$$SO_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t. } A \cdot A^T = I \text{ and } \det(A) = 1 \}$$

* Lemma 1.22

Let A, B, C be square matrices and x a column vector s.t

$$Ax = B \quad \text{--- (i)}$$

$$CAx = CB \quad \text{--- (ii)}$$

If C is invertible $\Rightarrow$ solution space of (i) = solution space of (ii)

If C is non invertible $\Rightarrow$ solution space of (i) is a subspace of the solution space of (ii)

Groups

* Defn 2.1 Law of composition on set S is $f: S \times S \rightarrow S$

* Def 2.2 $e \in S$ is called identity w.r.t a law of composition if $e \cdot a = a \cdot e = a \quad \forall a \in S$
 Caution: A set might not have an identity element w.r.t a law of composition

* Defn 2.3 Transposition is an element of S_n that switches only two elements

* Defn 2.4 Group: A set G together with a law of composition $(\cdot)$ s.t

① $(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \forall a, b, c \in G$

② $\exists e \in G$ s.t $e \cdot a = a \cdot e = a \quad \forall a \in G$

③ if $a \in G \exists a^{-1} \in G$ s.t $a \cdot a^{-1} = a^{-1} \cdot a = e$

* Example 2.5 S_3 : Symmetry group of 3 elements

$x = (123) \quad y = (12)$

$x^3 = 1 \quad y^2 = 1 \quad yx = x^2y$

$S_3 = \{1, x, x^2, y, xy, x^2y\}$

generators

defining relations

* Defn 2.6 Subgroup H of group G , is a non empty subset s.t

① if $a, b \in H$ and $c = a \cdot b \Rightarrow c \in H$

② if $a \in H \Rightarrow a^{-1} \in H$

* Example 2.7

$$\begin{matrix} & C_n & \\ SO_n & \subset & C_n \\ & \subset & S_n \end{matrix}$$

* Defn 2.8 $H \leq G$ is called normal subgroup, i.e $H \trianglelefteq G$

if $\forall g \in G$ and $h \in H \quad ghg^{-1} \in H$

* Defn 2.9 Order of group G is the number of unique elements in G .

* Defn 2.10 Let U be the subset of group G . Then the group generated by U is defined as the smallest subgroup H of G that contains all elements of U

* Defn 2.11 Cyclic subgroup: The group generated by a single element x of a group
 $\langle x \rangle := \{ \dots, x^2, x^1, e, x, x^2, \dots \}$
 Order of $x :=$ Order of $\langle x \rangle$

* Defn 2.12 Group Homomorphism: $\varphi: G \longrightarrow G'$ s.t.
 $\varphi(a.b) = \varphi(a). \varphi(b) \quad \forall a, b \in G$

Image of $\varphi := \{ \varphi(g) : g \in G \}$
 kernel of $\varphi := \{ g \in G : \varphi(g) = e' \}$

* Lemma 2.13 (i) $\varphi(e) = e'$
 (ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$
 (iii) if $H \subset G \implies \varphi(H) \subset G'$

* Lemma 2.14 (i) kernel is a normal subgroup of G
 (ii) Image is a subgroup of G'

* Lemma 2.15 A homomorphism φ is injective i.e one to one if kernel of φ is $\{e\}$

* Defn 2.16 Isomorphism: A bijective group homomorphism

* Defn 2.17 Automorphism: An isomorphism from group to itself

* Defn 2.18 $a, b \in G$ are called conjugates if $\exists g \in G$ s.t $b = g a g^{-1}$

* Defn 2.19

Let $H \leq G$ and $a \in G$ then

$$\text{Left coset of } a := aH = \{a \cdot h \in G : h \in H\}$$

↳ All the left cosets have the same size

* Theorem 2.20

Let $H \leq G$ then

$$[G] = [H] [G:H]$$

order of G order of H # of left cosets

Hint: $a \sim b$ if $\exists h \in H$ s.t. $a = b \cdot h$ is an equivalence relation $\Downarrow$

$$[G] = [H] [G:H] \quad \leftarrow \begin{array}{l} \text{Left cosets are non intersecting} \\ 2.19 \end{array}$$

* Corollary 2.21

Let $\varphi: G \rightarrow G'$ be a group homomorphism,

$$[G] = [\text{kernel}] [\text{Image}]$$

* Correspondence: Theorem 2.22 Let $\varphi: G \rightarrow G'$ be a surjective group homomorphism

$$(i) \quad \left[\begin{array}{l} \text{Subgroups of } G \text{ that} \\ \text{contain the kernel} \end{array} \right] \xleftrightarrow[\text{map}]{\text{bijective}} \left[\begin{array}{l} \text{Subgroups of} \\ G' \end{array} \right]$$

$$(ii) \quad \{ H \triangleleft G \text{ and } K \subset H \} \xleftrightarrow[\text{map}]{\text{bijective}} \{ H' \triangleleft G' \}$$

* Notes: surjection of φ is needed for

(a) the correspondence to be onto

(b) $\varphi(H) \triangleleft G'$ where $H \triangleleft G$ and $K \subset H$

* Defn 2.23

Quotient Group: Let $H \triangleleft G$, $\bar{a} := aH$ left coset of $a \in G$ then

$G/H := \{ \bar{a} : a \in G \}$ with the law of composition $\bar{a} \cdot \bar{b} = \overline{a \cdot b}$
has the group structure

Further $\varphi: G \rightarrow G/H$ where $\varphi(a) = \bar{a}$ is a group homomorphism

Hint: Prove that law of composition is valid for arbitrary representative element and then it carries over the group structure

* First
Isomorphism
Theorem 2.24

Let $\varphi: G \rightarrow G'$ is a surjective homomorphism and K be its kernel, then
 $\exists$ an isomorphism f b/w G/K and G'

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & G' \\ \pi \searrow & & \nearrow f \\ & G/K & \end{array}$$

$$f(\bar{a}) := \varphi(a)$$

* Lemma
2.25

If w_1 and w_2 are subgroups of G , then the following are also subgroups

$$w_1 \cap w_2 := \{g \in G : g \in w_1 \text{ and } g \in w_2\}$$

$$w_1 + w_2 := \{g \in G : g = a \cdot b \text{ or } g = b \cdot a \text{ where } a, b \in w_1 \cup w_2\}$$

* Usually $w_1 + w_2 \neq w_1 \cup w_2$

* Lemma
2.26

All the subgroups of $\mathbb{Z}^+$ (additive group of integers) have the form
 $n\mathbb{Z}$ where $n \in \mathbb{Z}$.

Hint: Use well ordering property of $\mathbb{Z}$

* GCD
Defn 2.27

If $a, b \in \mathbb{Z}$, then $d \in \mathbb{Z}$ is called gcd if

(i) $d > 0$

(ii) d divides a and b

(iii) if $c \in \mathbb{Z}$ divides a and b then c divides d

* Lemma 2.28

If $a, b, d \in \mathbb{Z}$ st d is gcd of a and $b \implies d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$

* Defn 2.29

$a, b \in \mathbb{Z}$ are called relatively prime if $\text{g.c.d}(a, b) = 1$

* Defn 2.30

Let $p \in \mathbb{Z}$ st if p divides $a \cdot b \implies p$ divides either a or b

* Lemma 2.31

$p \in \mathbb{Z}$ is prime $\iff \exists n \in \mathbb{Z}$ st $1 < n < p$ and n divides p .

* L.C.M If $a, b \in \mathbb{Z}$, then $l \in \mathbb{Z}$ is called gcd if

defn 2.32

(i) $l > 0$

(ii) a and b divides l

(iii) if a and b divides $c \in \mathbb{Z}$ then l divides c

* Lemma 2.33 If $a, b, l \in \mathbb{Z}$ s.t l is l.c.m of a and $b \implies l\mathbb{Z} = a\mathbb{Z} \cap b\mathbb{Z}$

* Lemma 2.34 $\text{g.c.d}(a, b) \times \text{l.c.m}(a, b) = a \cdot b$

Prove (i) $a \cdot b / d$ is divisible by l using 2.32

(ii) $l \cdot d$ is divisible by $a \cdot b$ using $d = ma + nb$

* Defn
2.48

Modular arithmetic

Congruence Relation $a \equiv b$ if $a - b$ is divisible by n

$f: \mathbb{Z}^+ \rightarrow \mathbb{Z}/n\mathbb{Z}$ is group homomorphism with $f(a+b) = f(a) + f(b)$

further (i) $f(a \cdot b) = f(a) \cdot f(b)$

(ii) $f(a) (f(b) + f(c)) = f(a) f(b) + f(a) f(c)$

Vector Space

- * Defn 3.1 A field F is a set together with two laws of composition, i.e. addition ⁽⁺⁾ and multiplication ^(*) st
- (i) $(F, +)$ is an abelian group with identity denoted by 0
 - (ii) $(F, \cdot)$ is an abelian group, where $F^* = \{F - \{0\}\}$. Identity is denoted by 1.
 - (iii) $\forall a, b, c \in F \quad a \cdot (b + c) = ab + ac$

- * Lemma 3.2 $F_p = \mathbb{Z}/p\mathbb{Z}$ where $p \in \mathbb{Z}$ is prime is a field.
- Hint: Proof by contradiction. Let $p = a \cdot b$ with $1 < a, b < p$
show that closure fails for F_p^*

- * Lemma 3.3 F_p^* is an group of order $p-1$ and is cyclic, i.e.
- $$F_p^* = \{1, \alpha, \alpha^2, \dots, \alpha^{p-2}\} = \{\bar{1}, \bar{\alpha}, \dots, \overline{p-1}\}$$
- The generator α of F_p^* is called primitive root modulo p .

- * Defn 3.4 The characteristic of a field is the order of $\langle 1 \rangle \subset (F, +)$

- * Lemma 3.5
- characteristic of a field

Finite

It is a prime number

Infinite

then it is called zero

Hint: Same as 3.2

- * Defn 3.6 A vector space V over a field F is a set together two laws, i.e. vector addition and scalar multiplication s.t
- (i) V is an abelian group with scalar addition. Identity element is 0_V
 - (ii) if $a \in F, v \in V$ then $a \cdot v \in V$
 - (iii) if $v \in V \quad 1 \cdot v = v$
 - (iv) if $u, v \in V$ and $a, b \in F \implies$

$$a(u+v) = au + av$$

$$(a+b)v = av + bv$$

$$a(b \cdot v) = (a \cdot b)v$$

- * Lemma 3.7 let $v \in V, 0_F \in F \implies 0_F \cdot v = 0_V$

* Defn 3.8

A subspace of a vector space V over field F is a non empty subset s.t

- ① if $u, v \in V$ then $u+v \in V$
- ② if $a \in F, v \in V$ then $a \cdot v \in V$

* Lemma 3.9

If W_1 and W_2 are subspaces of V then

- ① $W_1 \cap W_2$ is a subspace of V
- ② $W_1 + W_2$ is a subspace of V , where $W_1 + W_2$ is the group generated by $W_1 \cup W_2$ (2.25)

* Defn 3.10

Ordered Set $\begin{cases} \rightarrow \text{Order of elements matter} \\ \rightarrow \text{Repetition of elements is not allowed.} \end{cases}$

* Defn 3.11

Span of an ordered set $S = \{v_1, v_2, \dots\}$ with $v_i \in V$ is the subspace generated by linear combinations of v_i 's, i.e.

$$\text{Span}(S) := \left\{ \sum_i a_i v_i \text{ where } a_i \in F, v_i \in V \right\}$$

* Defn 3.12

A vector space V is called finite dimensional if $\exists$ a finite set S s.t
 $V = \text{Span}(S)$

* Defn 3.13

An ordered set $S = \{v_1, v_2, \dots, v_n\}$ with $v_i \in V$ is called linearly independent if
 $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0_V \iff a_i = 0 \text{ where } 1 \leq i \leq n$

* Defn 3.14

An empty set $\emptyset$ is linearly independent
 $\text{Span}(\emptyset) = \{0_V\}$

* Defn 3.15

A basis B of vector space V is an subset s.t

- ① B is linearly independent
- ② $\text{Span}(B) = V$

* Defn 2.16 The dimension of a finite dimensional vector space is defined to be number of vectors in its basis

* Lemma 3.17 Let L be a linearly independent set of finite dimensional V , S be a set of V s.t. $\text{Span}(S) = V$ then $\dim(\text{Span}(L)) \leq \dim(\text{Span}(S))$

hint: Proof by contradiction $\dim(\text{Span}(L)) > \dim(\text{Span}(S))$

* Lemma 3.18 Let V be a finite dimensional vector space and $[B]$ and $[B]'$ are its two basis associated by $[B]' = [B]P$ $\begin{matrix} \xrightarrow{\quad} \\ \downarrow \end{matrix}$ $\begin{matrix} P \text{ is invertible} \\ \text{base change matrix} \end{matrix}$

If $v \in V$ then $\alpha = [B]'x' = [B]x$ $\xrightarrow{\quad}$ $\begin{matrix} \text{coordinate vector of } v \text{ w.r.t } [B] \end{matrix}$

$x = Px'$

* Defn 3.19 The two subspaces U_1 and U_2 of V are called linearly independent if $U_1 \cap U_2 = \{0\}$

* Defn 3.20 A vector space V is called direct sum of subspace $U_1, U_2, \dots, U_k$ if

- ① $U_1, U_2, \dots, U_k$ are linearly independent
- ② $V = U_1 + U_2 + \dots + U_k$

It is denoted by $V = U_1 \oplus U_2 \oplus \dots \oplus U_k$

* Lemma 3.21 If W_1 and W_2 are subspace of V then $\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$

Linear Transformation

* Defn 4.1 A linear transformation T is a map from vector space V to vector space W , both over the same field F s.t

$$\textcircled{1} T(u_1 + u_2) = T(u_1) + T(u_2) \quad \forall u_1, u_2 \in V$$

$$\textcircled{2} T(av) = aT(v) \quad \forall v \in V, \forall a \in F$$

* Defn 4.2
Kernel of $T := \{u \in V : T(u) = 0_W\}$
Image of $T := \{T(u) : u \in V\}$

* Lemma 4.3
Kernel is a subspace of V
Image is a subspace of W

* Lemma 4.4
 $\dim(\text{Kernel}) + \dim(\text{Image}) = \dim(V)$
nullity rank

Hint $[B] := \{ \overbrace{a_1, a_2, \dots, a_k}^{\text{basis of kernel}}, b_1, \dots, b_m \}$
Show $\{T(b_1), T(b_2), \dots, T(b_m)\}$ is a basis of Image

* Lemma 4.5
Let $T: V \rightarrow W$ be a linear transformation, P a subspace of V
then $T(P)$ is a subspace of Image

* Defn 4.6
Let $T: V \rightarrow W$ be a linear transformation, $[B], [C]$ be the basis of V and W respectively
Then the matrix of linear transformation T w.r.t $[B]$ and $[C]$ is a matrix A s.t

$$T([B]) = [C]A$$

Lemma
If $x \in F^n, y \in F^m$ be the coordinate vectors of $u \in V$ and $T(u) \in W$ w.r.t $[B]$ and $[C]$ respectively
then $y = Ax$

Lemma
If $[B']$ and $[C']$ are another sets of basis of V and W s.t
 $[B'] = [B]P \quad [C'] = [C]Q$

then A' , the matrix of linear transformation T w.r.t $[B']$ and $[C']$ is given by
 $A' = Q^{-1}AP$

* Fig 47